

ASE379L HWS

1.

1.1 $S: \{I, F, L, R\}$

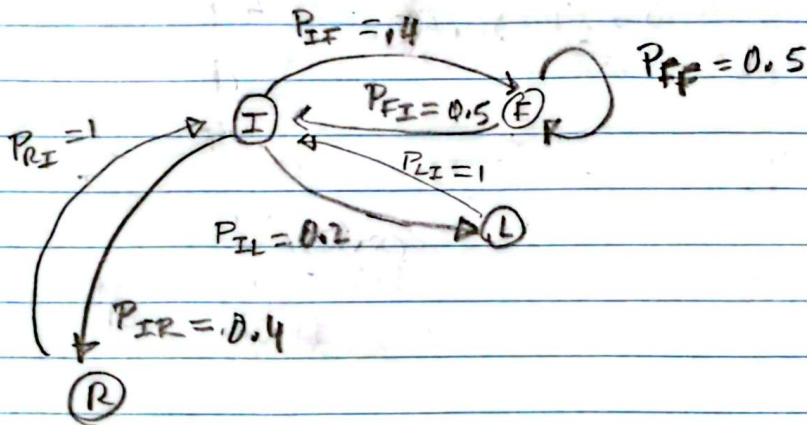
I: at intersection $\rightarrow$ stops

F: forward

L: left

R: right

1.2



$P_{FI} = P_{LI} = P_{RI} = 1$ since the car always goes to the next intersection I $\rightarrow$ this is a deterministic step

$$P_{IF} = .4$$

$$P_{IL} = 0.2$$

$$P_{IR} = 0.4$$

$$P_{II} = 0.2$$

$P_{FF} = 0.5$ (if it does not stop, we do not turn/move to another state).

$P_{FI} = 0.5$ $\rightarrow$ equal probability between stopping and continuing forward

1.3 Transition Kernel P:

$$\text{let state} = \begin{Bmatrix} I \\ F \\ L \\ R \end{Bmatrix} = \vec{u}$$

$$\text{then: } P = \begin{bmatrix} 0 & 0.4 & 0.2 & 0.4 \\ 0.5 & 0.5 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

(rows sum to 1)

$$1.4 \quad P_I = P_F = P_L = P_R \quad \text{at } t=0. \\ = 0.25.$$

$$\text{So, } p_0 = \begin{bmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 0.25 \end{bmatrix}$$

at step 10, we have $(P)^{10} p_0$:

$$p_{10} = (P)^{10} p_0 \approx \begin{bmatrix} 0.52 & 0.30 & 0.0589 & 0.117 \\ 0.38 & 0.34 & 0.092 & 0.18 \\ 0.24 & 0.37 & 0.112 & 0.22 \\ 0.24 & 0.37 & 0.112 & 0.22 \end{bmatrix}$$

$$= [0.37, 0.346, 0.09, 0.19]$$

2. For a diagonal transition matrix, Transitions between states cannot occur.

Each row must sum to 1, since, from a given state, the probability of moving to any other possible state (including remaining at the current state) must sum to 100%.

As such $A_i = 1 \quad \forall i \in [1, N]$

So, $P = I_N$ (identity of size $N \times N$)

and application of P on any state vector

p_0 will cause $p_1 = p_0 P = p_0$.

↑

Step 0

The chain is thus reducible, with n independent states